KATHRYN B. NEWHART

kathryn.newhart@oregonstate.edu

PROFESSIONAL EXPERIENCE

Oregon State University June 2024 - Present
Assistant Professor of Environmental Engineering *Corvallis,OR*

United States Military Academy June 2021 - June 2024
Assistant Professor of Environmental Engineering *West Point, NY*

Metro Wastewater Reclamation District March 2020 - May 2021
Technology & Innovation Engineer Associate *Denver, CO*

EDUCATION

Doctor of Philosophy 2018 - 2020
Civil and Environmental Engineering *Colorado School of Mines, Golden, CO*

· Dissertation: "Data-driven process control of municipal wastewater treatment"
· Advisors: Prof. Tzahi Cath and Prof. Amanda Hering (Baylor University)

Master of Science 2016 - 2018
Civil and Environmental Engineering *Colorado School of Mines, Golden, CO*

Bachelor of Science 2013 - 2016
Environmental Engineering *Colorado School of Mines, Golden, CO*

GRANTS AND AWARDS

"Integrated Computer Vision Intelligence System for Activated Sludge Microscopy and Process Optimization," Water Research Foundation, Tailored Collaboration Program, 5389, Awarded 2026. Total award $300,000. OSU award $175,000. Principal Investigator.

"Digital Water for DOD Installations: Demonstration and Deployment Guidance for Leveraging Big Data at Water and Wastewater Utilities," U.S. Department of Defense, Environmental Security Technology Certification Program, Awarded 2025. Total award $1,010,064. OSU award $589,914. Principal Investigator.

"Increasing Installation Energy Resilience through Energy Producing Resource Recovery Facilities," U.S. Department of Defense, Environmental Security Technology Certification Program, Awarded 2025. Total award $1,566,825. OSU award $25,050. co-Principal Investigator.

"Unlocking the Nationwide Potential of Water Reuse," U.S. Environmental Protection Agency, National Priorities: Water Innovation, Science, and Engagement to Advance Water Reuse. EPA-G2021-ORD&EI. Awarded 2023. Total award $4,000,000. Research Partner at West Point.

"Data-driven Fault Detection and Process Control for Potable Reuse with Reverse Osmosis," National Alliance for Water Innovation, Autonomous Water and Precision Separations. NAWI-2-2021. Awarded 2022. Total award $1,034,000. co-Principal Investigator.

"Crossing the Finish Line: Integration of Data-Driven Process Control for Maximization of Energy and Resource Efficiency in Advanced Water Resource Recovery Facilities," U.S. Department of Energy, Research and Development for Advanced Water Resource Recovery Systems. DE-FOA-0002336. Awarded 2021. Total award $2,400,000. co-Principal Investigator.

"Advancing Energy Resilience through Demonstrating, Characterizing, and Modeling Anaerobic Co-digestion of Organic Wastes for Widespread Implementation on DOD Installations." U.S. Department of Defense, Environmental Security Technology Certification Program. Total award $925,000. co-Principal Investigator.

PUBLICATIONS

15. Grimm, T.R., **Newhart, K.B.** and Hering, A.S.,"A Robust Self-starting Bayesian Approach for Multivariate Phase II Monitoring," *Quality and Reliability Engineering International*, 2026, *in press*, DOI: 10.1002/qre.70213

14. **Newhart, K.B.**, Thompson, K.A., Branch, A., Saleson, A., Safarik, J., Plumlee, M.H., "NDMA soft-sensors for potable reuse: A model development study," *Water Research*, 2025, 287, 124451, DOI: 10.1016/j.watres.2025.124451

13. Black, A., **Newhart, K.**, Linvill, C., Pytlar, A., Galaitsi, S., Fairfield;, C., Wait, M., Bennett, E., Butkus, M., "A linguistic analysis of energy terminology in the wastewater literature," *Water Science & Technology*, 2025, 91, 6, 671-683, DOI: 10.2166/wst.2025.036

12. Grimm, T.R., **Newhart, K.B.** and Hering, A.S.,"Nonparametric Threshold Estimation of Autocorrelated Statistics in Multivariate Statistical Process Monitoring," *Journal of Chemometrics*, 2025, 39, 2, DOI: 10.1002/cem.70004

11. Grimm, T.R., Branch, A., Thompson, K.A., Salevon, A., Zhao, J., Johnson, D., Hering, A.S., **Newhart, K.B.**, "Long-term Statistical Process Monitoring of an Ultrafiltration Water Treatment Process," *ACS ES&T Engineering*, 2024, DOI: 10.1021/acsestengg.4c00042

10. **Newhart, K.B.**, Klanderman, M.C., Hering, A.S., Cath, T.Y., "A holistic evaluation of multivariate statistical process monitoring in a biological and membrane treatment system," *ACS ES&T Water*, 2023, DOI: 10.1021/acsestwater.3c00058

9. Martin, M., Goethals, P., **Newhart, K.**, Rhodes, E., Vogel, J., Stevenson, B., "Optimization of sewage sampling for wastewater-based epidemiology through stochastic modeling." *Journal of Engineering and Applied Science*, 2023, 70, 11.

8. **Newhart, K.B.**, Pfluger, A.R., Butkus, M.A., "The Green Escape Room: Part 2 – Teaching Students Professional Engineering Ethics by Applying Environmental Engineering Principles and Deciphering Clues and Puzzles." Paper presented at *2022 ASEE Annual Conference & Exposition*, Minneapolis, MN, 2022.

7. **Newhart, K.B.**, Hering, A.S., Cath, T.Y., "Data science tools to enable decarbonized water and wastewater treatment systems." *Pathways to Water Sector Decarbonization, Carbon Capture and Utilization*, edited by Z. Jason Ren and Krishna Pagilla, IWA Publishing, 2022.

6. **Newhart, K.B.**, Goldman-Torres, J., Wisdom, B. Freedman, D., Hering, A.S., Cath, T.Y., "Real-time dose control of peracetic acid disinfection in municipal wastewater treatment," *ACS ES&T Water*, 2021, 1, 2, 328–338

5. **Newhart, K.B.**, Marks, C.A., Rauch-Williams, T., Cath, T.Y., Hering, A.S. "Hybrid statistical-machine learning ammonia forecasting in continuous activated sludge treatment for improved process control," *Journal of Water Process Engineering*, 2020, 37, 101389

4. Klanderman, M., **Newhart, K.B.**, Cath. T.Y., Hering, A.S., "Fault isolation for a complex decentralized wastewater treatment facility," *Journal of the Royal Statistical Society*, Series C., 2020, 69, 931-951.

3. **Newhart, K.B.**., Holloway, R.W., Hering, A.S., Cath, T.Y., "Data-driven performance analyses of wastewater treatment plants: A review," *Water Research*, 2019, 157, 498-513

2. Odom, G.J., **Newhart, K.B.**, Cath, T.Y., Hering, A.S., "Multi-state multivariate statistical process control," *Applied Stochastic Models in Business and Industry*, 2018, 34(6), 880-892

1. Bell, E.A., Poynor, T.E., **Newhart, K.B.**, Regnery, J., Coday, B.D., Cath, T.Y., "Produced water treatment using forward osmosis membranes: evaluation of extended-time performance and fouling," *Journal of Membrane Science*, 2017, 525, 77-88.

WORKSHOPS

"Planning Your Data and Analytics Roadmap" *WEFTEC*, September 28, 2025, Chicago, IL, Co-organizer

"An Introduction to Machine Learning Tools for Solving Environmental Challenges" *AEESP*, May 2025, Duke University, Organizer

"Doing More, with Less: Implementing Machine Learning Process Controls at WRRFs" *WEFTEC*, October 6, 2024, New Orleans, LA, Co-presenter

"Defining Pathways for Solving Environmental Challenges using Machine Learning" *AEESP Annual Conference*, May 2023, Organizer

"A Convergence of WRF Machine Learning Based Controller Implementation and Research" *WEF/IWA Innovations in Process Engineering*, June 6, 2023, Facilitator and presenter

"Visualization, Analysis, and Modeling in R for the Water Professional" MoWaTER PRO: Data Science Workshop, 2021 & 2022, Organizer

"Machine Learning in the Water Industry" *WEF/IWA Innovations in Process Engineering*, June 8, 2021, Organizer

"Understanding and Embracing Machine Learning, Artificial Intelligence and Predictive Analytics," *AWWA/SWAN International Smart Water Symposium*, November 10, 2020, Facilitator and presenter

"Data Research Advances Water Industry," *NSF Mid-scale Research Infrastructure Workshop for Intelligent Water Systems*, August 25, 2020, Virtual, Facilitator and presenter

INVITED TALKS

"AI & Digital Tools in Water Research" *National Alliance on Water Innovation Next Gen Seminar Series*, January 28, 2026, Webcast, Panelist

"Data-driven modeling in the water sector and the paradigm shift in predictive methods" *IWA Modeling and Integrated Assessment Seminar Series*, December 10, 2025, Webcast, Presenter

"Bridging Infrastructure and Intelligence: AI-Enhanced Water Resource Recovery," *Marquette University*, November 12, 2025

"Emerging Challenges and Research Frontiers for Artificial Intelligence in Water and Wastewater Treatment," *National Academy of Science: Water Science and Technology Board*, Frontier Applications of AI and Water Management, October 30, 2025, Presenter and Panelist

"From Data to Decisions: Why Machine Learning is Difficult but Essential for 21st Century Water Treatment Systems," *Oregon State University*, October 24, 2025

"GenAI in Capstone," *AEESP Education Committee*, Incorporating AI into the environmental engineering and science curriculum, Presenter, May 9, 2025, URL

"Digital Dreams: AI & the Water Sector," New York's Water Event, *NYAWWA*, April 15, 2025, Keynote Address

"Implementing Machine Learning Process Controls at Water Resource Recovery Facilities" *Water Research Foundation*, March 18, 2025, Webcast, Presenter

"Data-Driven Fault Protection and Process Control for Potable Reuse with Reverse Osmosis," *National Alliance for Water Innovation*, January 22, 2025, Webcast, Presenter

"Digital Water: Leveraging Statistical and Machine Learning in Water Treatment Research," *Univertsity of Michigan*, July 2024

"Machine Learning to the Rescue: Mining data to optimize water reclamation disinfection," *Water Environment Federation*, June 13, 2024, Webcast, Presenter

"Process Modeling and Machine Learning – Selecting the Right Tool(s) for Troubleshooting and Optimization," *Water Environment Federation*, November 30, 2023, Webcast, Presenter

"Artificial Intelligence in the Water Industry," *Orange County Water District*, August 1, 2023, Webcast, Presenter, URL

"Understanding and Embracing Machine Learning, Artificial Intelligence and Predictive Analytics" *Metropolitan Water Reclamation District of Greater Chicago*, June 30, 2023, Seminar

"A Hypothetical – Potable Reuse Moves Towards Artificial Intelligence," *36th Annual WateReuse Symposium*, March 1, 2021, Panelist

SELECT CONFERENCE PRESENTATIONS

"CURATE: Leveraging LLM Agents to Compose, Catalog, and Deploy Reproducible Workflows" *Workshop on Workflows in Support of Large-Scale Science (WORKS)*, 2026, Chicago, IL

"From Machine Learning to Agentic AI: Exploring the Journey for Utilities and their Workforce" *WEFTEC*, 2025, Chicago, IL

"Prediction of Post-Secondary E. coli for Disinfection Control: Application of statistical and machine learning algorithms." *WEF/IWA Innovations in Process Engineering*, June 9, 2023, Portland, OR

"Predictive Control in Wastewater Treatment Facilities Using Simple Statistical Models," *South Platte Coalition for Urban River Evaluation: Confluence at the Confluence*, Oct 15, 2019, Englewood, CO

"Energy Reduction in Municipal Wastewater Treatment," *Colorado Industrial Pretreatment Coordinators Association Fall Conference*, Oct 18, 2019, Black Hawk, CO

"Predictive Modeling and Performance Assessment of Ammonia-Based Aeration Control," *WEFTEC*, Sept 23, 2019, Chicago, IL

"A Utility Perspective: Practical Considerations of Operating and Advancing Ammonia-Based Aeration Control," July 10, 2019, *RMWEA Innovation Seminar*, Denver, CO

"Fault Detection Using PCA at a Municipal Wastewater Treatment Facility," July 30, 2019, *Joint Statistical Meeting*, Denver, CO

"Performance Evaluation of a Sequencing Batch Membrane Bioreactor Using Principal Component Analysis," *Annual WateReuse Symposium*, Sept 11, 2017, Phoenix, AZ

"Use of Principal Component Analysis for Early-Fault Detection in a Pilot-Scale Biological Wastewater Treatment System," *Quality and Productivity Research Conference*, June 14, 2017, Storrs, CT

NON-REFEREED PUBLICATIONS

Weintraut, Z., **Newhart, K.**, Thompson, K., Roostaei, J., "Are you ready for big data? A checklist for readiness for data analytics in water utilities," *Journal AWWA*, 2022, 114, 10, 78-82

Newhart, K.B. & Avila, I., "NDMA: relevance and regulatory status for drinking water facilities," *Rocky Mountain Water*, November 2017

AWARDS

ASEE Environmental Engineering Division Early Career Award, 2022

ACS Publications Peer Reviewer, Certificate of Recognition, 2022

WEF/WRF LIFT Intelligent Water System Challenge, 1st place, 2019

AWRA-Colorado Rich Herbert Memorial Scholarship, 2019

LEADERSHIP

Chair, National Water Research Institute (NWRI) Independent Expert Panel, supporting *Data-Driven Fault Detection and Process Control for Potable Reuse with Reverse Osmosis Project*

President, NSF ReNUWIt Engineering Research Center Student Leadership Committee, 2018 – 2019

President, CSM Campus Chapter of the Rocky Mountain Section of the American Water Works Association (RMSAWWA)/Rocky Mountain Water Environment Association (RMWEA), 2018 – 2019

Co-Chair, 15th Annual RMSAWWA/RMWEA Joint Student Conference, 2018

SERVICE

Institution (OSU)

- Program Safety Committee, September 2024– present

Professional

- Reviewer, National Alliance for Water Innovation, Industrial Water Reuse, NAWI-4-2026, October 2026

- Early Career Board, ACS Environmental Science & Technology Engineering, January 2024 – present
- Editorial Advisory Team, ACS Environmental Science & Technology Water, June 2022 – present
- Referee, ACS Environmental Science & Technology Engineering; Environmental Science: Water Research & Technology; Resources, Conservation & Recycling; Water Environment Research; Water Research (IWA); Journal of Water & Health; Springer Nature: Scientific Reports
- Technology Reviewer, Water Research Foundation TechLink, 2022
- Member, AWWA Water Science & Research Division, Information Management & Technology, 2021 – 2023

IN THE NEWS

Newhart, K. B., Marks, C., Rauch-Williams, T., Cath, T. Y., Hering, A. S. (2020) "Boulder tests its waters with predictive aeration control," Advances in Water Research, 30: 25–28. URL.

CERTIFICATIONS

Wastewater Operator, Class D, Colorado, 2016-2024

Fundamentals of Engineering (FE), Environmental, Colorado, NCEES ID 16-475-7

TEACHING

Institution	*Course*	*Title (Credit Hours)*	*Terms*
OSU	ENVE 490[1]	Environmental Engineering Design (3)	1
OSU	ENVE 415[1]	Environmental Engineering Laboratory (3)	1
USMA	EV201[1]	Introduction to Environmental Engineering (3)	2
USMA	EV350[2]	Environmental Engineering Technologies (3)	1
USMA	EV401[1]	Physical and Chemical Treatment (3.5)	3
USMA	EV450[2]	Environmental Engineering for Sustainable Development (3)	3
USMA	EV490/491[2]	Environmental Engineering Design (Capstone) (3)	5
USMA	XE365[3]	Advanced Experimental Methods & Data Processing (3)	1
RRCC	WQM42[1]	Water Data Management & Analysis (3)	1
CSM	CEE 470/570[3]	Unit Processes for Water and Wastewater Treatment (3)	3
CSM	CEE 471/571[3]	Advanced Water Treatment and Reclamation (3)	1
CSM	CEE 330[3]	Field Session for Environmental Engineering (3)	3

[1] Course director, [2] Instructor, [3] TA / Guest lecturer

ADVISEES

Primary Advisor

Perry Wilson, MS, Environmental Engineering, 2024-Present

Committee Member

Matthew Latham, PhD, Environmental Engineering, 2023-Present

Sampson Achagwe Antwi, PhD, Environmental Engineering, 2024-Present

Sofia Zook, PhD, Environmental Engineering, Colorado School of Mines, 2024-Present

Taylor Grimm, PhD, Statistical Science, Baylor University, 2021-2024